Sum and Difference Formulas

ANSWER KEY

Exact values with sum and difference formulas

1 Use a sum or difference formula to find the exact value of $\cos 75°$.

$\cos 75° = \cos(45° + 30°) = \cos 45° \cos 30° - \sin 45° \sin 30° = \frac{\sqrt{6} - \sqrt{2}}{4}$.

2 Find the exact value of $\sin \frac{\pi}{12}$.

$\sin \frac{\pi}{12} = \sin\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \sin \frac{\pi}{3} \cos \frac{\pi}{4} - \cos \frac{\pi}{3} \sin \frac{\pi}{4} = \frac{\sqrt{6} - \sqrt{2}}{4}$.

3 Find the exact value of $\tan 105°$.

$\tan 105° = \tan(60° + 45°) = \frac{\sqrt{3} + 1}{1 - \sqrt{3}}$. Rationalizing: $\frac{(\sqrt{3} + 1)^2}{(1 - \sqrt{3})(1 + \sqrt{3})} = \frac{4 + 2\sqrt{3}}{-2} = -2 - \sqrt{3}$.

Working with given values

4 Angles A and B are both in Quadrant I with $\sin A = \frac{4}{5}$ and $\cos B = \frac{5}{13}$. Find:

a $\sin(A + B)$ $\cos A = \frac{3}{5}$, $\sin B = \frac{12}{13}$, so $\sin(A + B) = \frac{4}{5} \cdot \frac{5}{13} + \frac{3}{5} \cdot \frac{12}{13} = \frac{56}{65}$.

b $\cos(A + B)$ $\cos(A + B) = \frac{3}{5} \cdot \frac{5}{13} - \frac{4}{5} \cdot \frac{12}{13} = -\frac{33}{65}$.

c the quadrant of $A + B$ Sine positive, cosine negative: Quadrant II.

5 Use a sum or difference formula to simplify:

a $\sin\left(x + \frac{\pi}{2}\right) \quad = \cos x$

b $\cos(x - \pi) \quad = -\cos x$

6 Angles A and B are both in Quadrant II with $\sin A = \frac{8}{17}$ and $\sin B = \frac{3}{5}$. Find $\cos(A - B)$ exactly.

In QII, $\cos A = -\frac{15}{17}$ and $\cos B = -\frac{4}{5}$.

$\cos(A - B) = \cos A \cos B + \sin A \sin B = \left(-\frac{15}{17}\right)\left(-\frac{4}{5}\right) + \frac{8}{17} \cdot \frac{3}{5} = \frac{60}{85} + \frac{24}{85} = \frac{84}{85}$.

Verifying identities using sum and difference formulas

7 Verify the identity $\cos(x + y) + \cos(x - y) = 2\cos x \cos y$.

$\cos(x + y) = \cos x \cos y - \sin x \sin y$ and $\cos(x - y) = \cos x \cos y + \sin x \sin y$. Adding: the $\sin x \sin y$ terms cancel, leaving $2\cos x \cos y$.

8 Verify the identity $\sin(x + y) - \sin(x - y) = 2\cos x \sin y$.

$\sin(x + y) = \sin x \cos y + \cos x \sin y$ and $\sin(x - y) = \sin x \cos y - \cos x \sin y$. Subtracting: the $\sin x \cos y$ terms cancel, leaving $2\cos x \sin y$.

9 Show that $\tan\left(x + \frac{\pi}{4}\right) = \frac{1 + \tan x}{1 - \tan x}$ using the tangent sum formula $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$.

With $B = \frac{\pi}{4}$, $\tan B = 1$: $\tan\left(x + \frac{\pi}{4}\right) = \frac{\tan x + 1}{1 - \tan x \cdot 1} = \frac{1 + \tan x}{1 - \tan x}$.

Applications

10 Two musical tones combine to form a sound wave modeled by $y = \sin\left(x + \frac{\pi}{6}\right)$. Expand this expression using the sum formula, and evaluate the coefficient of $\cos x$ exactly.

$\sin\left(x + \frac{\pi}{6}\right) = \sin x \cos\frac{\pi}{6} + \cos x \sin\frac{\pi}{6} = \frac{\sqrt{3}}{2}\sin x + \frac{1}{2}\cos x$. The coefficient of $\cos x$ is $\frac{1}{2}$.

Additional practice

11 Find the exact value of $\cos\frac{\pi}{12}$.

$\cos\frac{\pi}{12} = \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \cos\frac{\pi}{3}\cos\frac{\pi}{4} + \sin\frac{\pi}{3}\sin\frac{\pi}{4} = \frac{\sqrt{6}+\sqrt{2}}{4}$.

12 Angles A and B satisfy $\tan A = \frac{3}{4}$ (QI) and $\tan B = \frac{5}{12}$ (QI). Find $\tan(A+B)$ using the tangent sum formula.

$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{3/4 + 5/12}{1 - (3/4)(5/12)} = \frac{14/12}{1 - 15/48} = \frac{7/6}{33/48} = \frac{7}{6} \cdot \frac{48}{33} = \frac{56}{33}$.

13 Simplify $\cos\left(\frac{\pi}{2} - x\right)\cos x + \sin\left(\frac{\pi}{2} - x\right)\sin x$ using the cosine difference formula in reverse.

This matches the pattern $\cos A \cos B + \sin A \sin B = \cos(A - B)$ with $A = \frac{\pi}{2} - x$, $B = x$: $= \cos\left(\frac{\pi}{2} - 2x\right)$.

A worked derivation

14 Starting from the cosine difference formula $\cos(A-B) = \cos A \cos B + \sin A \sin B$, derive the cosine sum formula $\cos(A+B)$ by substituting $B \to -B$.

$\cos(A+B) = \cos(A - (-B)) = \cos A \cos(-B) + \sin A \sin(-B)$. Since cosine is even and sine is odd: $= \cos A \cos B - \sin A \sin B$.

15 Find the exact value of $\sin 15° \cos 15°$ using a sum/difference-related double angle shortcut (recall $\sin(2\theta) = 2\sin\theta\cos\theta$).

$\sin 15° \cos 15° = \frac{1}{2}(2\sin 15° \cos 15°) = \frac{1}{2}\sin 30° = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Sum and difference formulas with tables of values

16 Given $\sin A = \frac{7}{25}$ (QI) and $\cos B = \frac{4}{5}$ (QI), find $\sin(A-B)$ exactly.

$\cos A = \frac{24}{25}$, $\sin B = \frac{3}{5}$. $\sin(A-B) = \sin A \cos B - \cos A \sin B = \frac{7}{25} \cdot \frac{4}{5} - \frac{24}{25} \cdot \frac{3}{5} = \frac{28}{125} - \frac{72}{125} = -\frac{44}{125}$.

17 Given $\tan A = 2$ (QI) and $\tan B = 3$ (QI), find $\tan(A-B)$ using the tangent difference formula.

$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} = \frac{2-3}{1+6} = -\frac{1}{7}$.

Expanding a general sinusoidal expression

18 Expand $\cos\left(x-\frac{\pi}{3}\right)$ fully using the difference formula, leaving the answer in terms of $\sin x$ and $\cos x$.

$\cos\left(x-\frac{\pi}{3}\right)=\cos x\cos\frac{\pi}{3}+\sin x\sin\frac{\pi}{3}=\frac{1}{2}\cos x+\frac{\sqrt{3}}{2}\sin x$.

A geometric interpretation

19 Explain, using the unit circle, why $\cos(A-B)$ (not $\cos A-\cos B$) correctly gives the cosine of the angle between two points at angles A and B on the unit circle.

The angle between two points on the unit circle at angles A and B is $|A-B|$, and by definition the cosine of that INCLUDED angle is $\cos(A-B)$ — a genuinely different quantity from the difference of the individual cosine values $\cos A-\cos B$, which has no direct geometric meaning as an angle's cosine.

Solving equations using sum and difference formulas

20 Solve $\sin\left(x+\frac{\pi}{6}\right)=\frac{1}{2}$ for $x\in[0,2\pi)$ by first expanding the left side.

Expand: $\sin x\cos\frac{\pi}{6}+\cos x\sin\frac{\pi}{6}=\frac{1}{2}\Rightarrow\frac{\sqrt{3}}{2}\sin x+\frac{1}{2}\cos x=\frac{1}{2}$. Alternatively, recognizing $\sin(\theta)=\frac{1}{2}$ directly: $x+\frac{\pi}{6}=\frac{\pi}{6}$ or $x+\frac{\pi}{6}=\frac{5\pi}{6}$, giving $x=0$ or $x=\frac{2\pi}{3}$.

21 Given that a signal is modeled by $y=\cos\left(3x-\frac{\pi}{4}\right)$, expand it using the difference formula and identify the coefficients of $\cos 3x$ and $\sin 3x$.

$\cos\left(3x-\frac{\pi}{4}\right)=\cos 3x\cos\frac{\pi}{4}+\sin 3x\sin\frac{\pi}{4}=\frac{\sqrt{2}}{2}\cos 3x+\frac{\sqrt{2}}{2}\sin 3x$. Coefficients: both $\frac{\sqrt{2}}{2}$.

22 Two AC voltage signals are given by $V_1(t)=\sin(\omega t)$ and $V_2(t)=\sin\left(\omega t+\frac{2\pi}{3}\right)$ (phase-shifted by $120°$, common in three-phase power systems). Expand $V_2(t)$ fully using the sum formula, in terms of $\sin(\omega t)$ and $\cos(\omega t)$.

$V_2(t)=\sin(\omega t)\cos\frac{2\pi}{3}+\cos(\omega t)\sin\frac{2\pi}{3}=-\frac{1}{2}\sin(\omega t)+\frac{\sqrt{3}}{2}\cos(\omega t)$.